MORPHISMS INTO PROJ FROM A LOCUS WHERE PSI GENERATES THE INVERTIBLE SHEAF

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Allowed standard prerequisites (editorial).

- Sections of twists on standard opens. lemma-proj-sheaves (same chapter, section-level display, not carded). Identifies the sections of the direct sum of twists over D_+(f) with the localization A_f.
- Converse construction. lemma-converse-construction (same chapter, judged non-self-contained this batch, not carded). Uniqueness input for the final agreement with the pair of that lemma.

Source reference. Original reference 00JM: focused-seeds/geometry/S9-cons-invertible-map-into-proj/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 01MP: sources/stacks-constructions/pinned-original-source.tex, source lines 1949–1949. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01N8: sources/stacks-constructions/pinned-original-source.tex, source lines 2666–2666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01IS: focused-seeds/geometry/S9-cons-invertible-map-into-proj/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let A be a graded ring. Set $X = Proj(A)$. Let T be a scheme. Let $\mathcal{L}$ be an invertible $\mathcal{O}_T$-module. Let $\psi : A \to \Gamma_*(T, \mathcal{L})$ be a homomorphism of graded rings. Set*

$$U(\psi) = \bigcup\nolimits_{f \in A_+ \text{ homogeneous}} T_{\psi(f)}$$

The morphism ψ induces a canonical morphism of schemes

$$r_{\mathcal{L},\psi} : U(\psi) \longrightarrow X$$

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together with a map of $\mathbf{Z}$*-graded* $\mathcal{O}_T$*-algebras*

$$\theta : r_{\mathcal{L},\psi}^* \left(\bigoplus\nolimits_{d\in\mathbf{Z}} \mathcal{O}_X(d) \right) \longrightarrow \bigoplus\nolimits_{d\in\mathbf{Z}} \mathcal{L}^{\otimes d}|_{U(\psi)}.$$

The triple $(U(\psi), r_{\mathcal{L},\psi}, \theta)$ *is characterized by the following properties:*

(1) *For* $f \in A_+$ *homogeneous we have* $r_{\mathcal{L},\psi}^{-1}(D_+(f)) = T_{\psi(f)}$.
(2) *For every* $d \geq 0$ *the diagram*

$$\begin{array}{ccc} A_d & \xrightarrow{\psi} & \Gamma(T, \mathcal{L}^{\otimes d}) \\ {\scriptstyle (01MP)}\downarrow & & \downarrow{\scriptstyle restrict} \\ \Gamma(X, \mathcal{O}_X(d)) & \xrightarrow{\theta} & \Gamma(U(\psi), \mathcal{L}^{\otimes d}) \end{array}$$

is commutative.

Moreover, for any $d \geq 1$ *and any open subscheme* $V \subset T$ *such that the sections in* $\psi(A_d)$ *generate* $\mathcal{L}^{\otimes d}|_V$ *the morphism* $r_{\mathcal{L},\psi}|_V$ *agrees with the morphism* $\varphi : V \to Proj(A)$ *and the map* $\theta|_V$ *agrees with the map* $\alpha : \varphi^*\mathcal{O}_X(d) \to \mathcal{L}^{\otimes d}|_V$ *where* (φ, α) *is the pair of Lemma 01N8 associated to* $\psi|_{A^{(d)}} : A^{(d)} \to \Gamma_*(V, \mathcal{L}^{\otimes d})$.

Proof. Suppose that we have two triples $(U, r : U \to X, \theta)$ and $(U', r' : U' \to X, \theta')$ satisfying (1) and (2). Property (1) implies that $U = U' = U(\psi)$ and that $r = r'$ as maps of underlying topological spaces, since the opens $D_+(f)$ form a basis for the topology on X, and since X is a sober topological space (see Algebra, Section 00JM and Schemes, Lemma 01IS). Let $f \in A_+$ be homogeneous. Note that $\Gamma(D_+(f), \bigoplus_{n\in\mathbf{Z}} \mathcal{O}_X(n)) = A_f$ as a $\mathbf{Z}$-graded algebra. Consider the two $\mathbf{Z}$-graded ring maps

$$\theta, \theta' : A_f \longrightarrow \Gamma(T_{\psi(f)}, \bigoplus \mathcal{L}^{\otimes n}).$$

We know that multiplication by f (resp. $\psi(f)$) is an isomorphism on the left (resp. right) hand side. We also know that $\theta(x/1) = \theta'(x/1) = \psi(x)|_{T_{\psi(f)}}$ by (2) for all $x \in A$. Hence we deduce easily that $\theta = \theta'$ as desired. Considering the degree 0 parts we deduce that $r^\sharp = (r')^\sharp$, i.e., that $r = r'$ as morphisms of schemes. This proves the uniqueness.

Now we come to existence. By the uniqueness just proved, it is enough to construct the pair (r, θ) locally on T. Hence we may assume that $T = \mathrm{Spec}(R)$ is affine, that $\mathcal{L} = \mathcal{O}_T$ and that for some $f \in A_+$ homogeneous we have $\psi(f)$ generates $\mathcal{O}_T = \mathcal{O}_T^{\otimes \deg(f)}$. In other words, $\psi(f) = u \in R^*$ is a unit. In this case the map ψ is a graded ring map

$$A \longrightarrow R[x] = \Gamma_*(T, \mathcal{O}_T)$$

which maps f to $ux^{\deg(f)}$. Clearly this extends (uniquely) to a $\mathbf{Z}$-graded ring map $\theta : A_f \to R[x, x^{-1}]$ by mapping $1/f$ to $u^{-1}x^{-\deg(f)}$. This map in degree zero gives the ring map $A_{(f)} \to R$ which gives the morphism $r : T = \mathrm{Spec}(R) \to \mathrm{Spec}(A_{(f)}) = D_+(f) \subset X$. Hence we have constructed (r, θ) in this special case.

Let us show the last statement of the lemma. According to Lemma 01N8 the morphism constructed there is the unique one such that the displayed diagram in its statement commutes. The commutativity of the diagram in the lemma implies the commutativity when restricted to V and $A^{(d)}$. Whence the result. □